National University of Computer and Emerging Sciences
Peshawar Campus

Artificial Intelligence (AI2002) / App. Artificial Intelligence (AI4007)

Sessional-II Exam

Date: 7th April 2025

Course Instructor(s)

Shahzeb Khan / Waqas Ali

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3

____________ ____________ ________________________

Roll No Section Student Signature

Do not write below this line

Attempt all the questions.

CLO 3: Apply methods of blind as well as informed search on problems related to programs.

Q1: The FAST Gaming Society recently organized a Game Night. The challenge was to transport freight from the storage area (S) to the event hosting rooms (G) as shown in the graph given in Figure 1. As an AI expert, the admin team seeks your assistance in determining the optimal path for transportation. Use the A* search algorithm to find the best route, utilizing all available information to design an effective solution. **[10]**

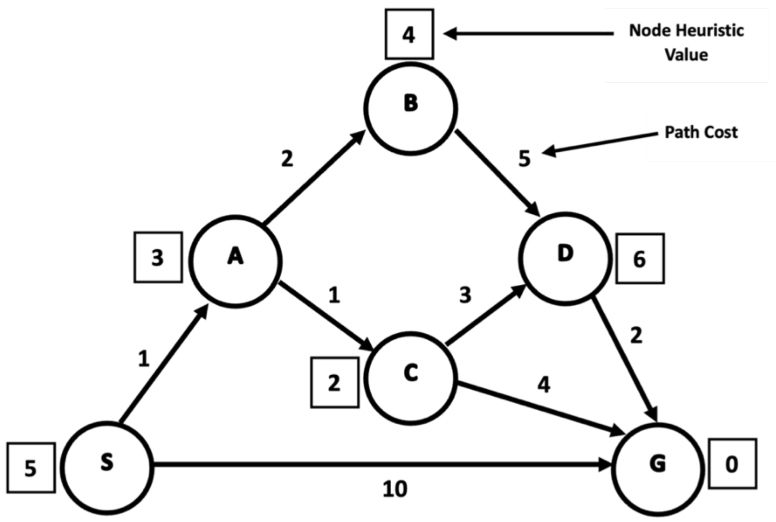


Figure 1

CLO 4: Summarize concepts and approaches in knowledge representation, planning, learning, robotics, and other AI areas.

Q2: A bank wants to predict whether a loan should be approved based on customer attributes using the **Naive Bayes Classifier**. The dataset shown in Table 1 contains past records of loan approvals with attributes: Income Level, Credit Score, Loan Purpose and Previous Default. **[10]**

Given a new applicant with attributes: Income Level = Medium, Credit Score = Fair, Loan Purpose = Education, and Previous Default = No, Use the **Naive Bayes Classifier** to determine whether the loan should be approved or not.

Table 1: Bank Loan Dataset

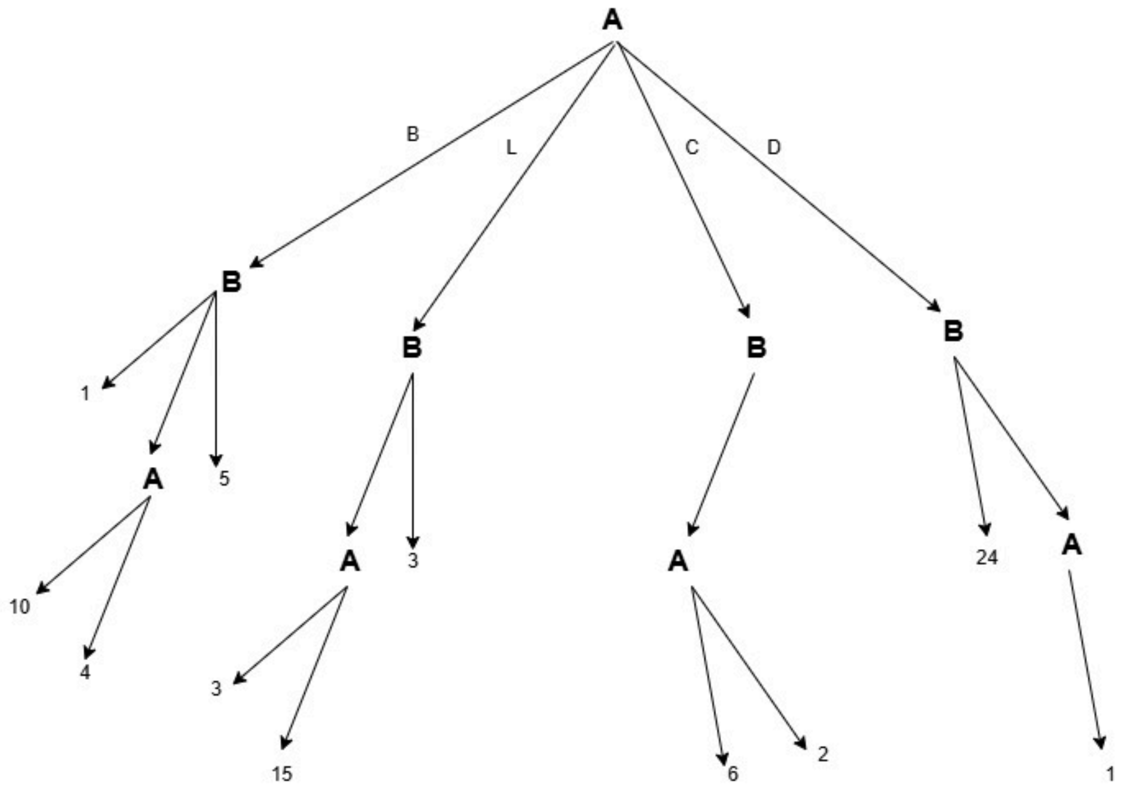
S.No.	Income Level	Credit Score	Loan Purpose	Previous Default	Loan Approved (Class)
1	High	Excellent	Business	No	Yes
2	Low	Poor	Personal	Yes	No
3	Medium	Fair	Education	No	Yes
4	Low	Fair	Business	Yes	No
5	High	Excellent	Personal	No	Yes
6	Medium	Poor	Education	Yes	No
7	High	Fair	Business	No	Yes
8	Low	Excellent	Personal	No	Yes

Note: Show all calculations. Incomplete work will lead to mark deductions.

CLO 5: Program AI applications by implementing the concepts learned in the course.

Q3: The FAST Sports Society organized Ramzan Sports Week, with cricket as the main event. The officials proposed a 22-step pitch length which was not accepted by the defending champions (Zalmi 11) as they wanted the pitch to be a maximum of 18 steps. Since the decision favored the organizers, Zalmi had four options: Boycott the tournament (B), Lobby against the tournament (L), Compete against all odds (C), or File a Dispute (D) against the sports society **[10]**

As an AI expert, design a Min-Max algorithm-based system to determine the best moves for Zalmi given the following Tree. **Note** that the root node is a max node.



---------------------- The End ----------------------